

Quantum Communications and Networks (QU.50013)

Exercise Set 1

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Problem 1

The benefits of applying quantum technology come not from communication speeds, but rather the usefulness of sharing quantum states itself. **Quantum key distribution** (QKD) is one of them: to ensure information-theoretic security, beyond the computational security provided by classical cryptography, we use the quantum mechanical principle that perfectly discriminating non-orthogonal states is impossible. Another is **quantum teleportation**, where unknown states can be sent far away, enabling distributed quantum computing systems. In both of these cases, qubits and ebits (entanglement) are the crucial quantum resources.

Problem 2

For a qubit in a pure state

$$|\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle,$$

measurement probabilities can be determined using the Born rule:

$$P(0) = |\langle 0|\psi\rangle|^2 = \frac{3}{4}, \quad P(1) = |\langle 1|\psi\rangle|^2 = \frac{1}{4}.$$

Problem 3

An unknown quantum state cannot be simply read, copied, and retransmitted as classical information could. This is called the *no-cloning theorem*, and it stems from the fact that quantum mechanics is inherently linear and unitary. The classical approach of measuring and copying fails because quantum measurement processes are destructive, i.e., measuring a system collapses its state into one of the measurement bases. Thus, quantum communication requires *quantum repeaters*, which help with retaining coherence in long-distance entanglements without amplifying signals.

Problem 4

No, local measurements on Alice's side cannot send any information to Bob. While entanglement allows for nonlocal *correlations* between separate measurements, it cannot create nonlocal *causations* between them. This can be readily seen in the reduced density matrix formulation, wherein Alice's local operations (unitary or measurement) does not affect Bob's reduced density matrix. Therefore, no matter what Alice does with her qubit, Bob's measurement outcomes are unaffected.

Problem 5

1. **Photon loss:** Photons are probabilistically absorbed or scattered by a realistic fiber optic cable, the likelihood of which increases with the distance travelled. Quantum repeaters mitigate this by breaking up the distance into shorter segments, swapping entanglements with each other and the endpoints.
2. **Decoherence:** Photons could interact with their environments, losing the coherent entangled state. While optical fibers are placed at ambient temperatures, repeaters operate in vacuum and cryogenic environments to reduce the probability of such decohering interactions.
3. **Probabilistic success:** Quantum repeater-based schemes often rely on random success events, which introduce temporal delays and could exacerbate the decoherence problem. One can use quantum memories with long lifetimes to hold onto a quantum state while waiting for the protocol to succeed.

Problem 6

The Bloch vector of a pure state $|\psi\rangle$ is found as

$$\vec{r} = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle).$$

Or equivalently, in the computational basis $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, one can find:

$$\vec{r} = (2\operatorname{Re}\{\alpha^*\beta\}, 2\operatorname{Im}\{\alpha^*\beta\}, |\alpha|^2 - |\beta|^2).$$

Therefore, we can find the corresponding Bloch vectors as:

- $|0\rangle \Rightarrow \vec{r} = (0, 0, 1)$
- $|1\rangle \Rightarrow \vec{r} = (0, 0, -1)$
- $|+\rangle \Rightarrow \vec{r} = (1, 0, 0)$

Problem 7

$$\begin{aligned} \rho &= |\psi\rangle\langle\psi| = \frac{1}{2} (|0\rangle\langle 0| - i|0\rangle\langle 1| + i|1\rangle\langle 0| + |1\rangle\langle 1|) \doteq \frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix} \\ &\Rightarrow \rho^2 = \rho \Rightarrow \operatorname{Tr}(\rho^2) = 1 \end{aligned}$$

This is consistent with the fact that $|\psi\rangle$ is a pure state.

Problem 8

$$\begin{aligned} \rho_+ &:= |+\rangle\langle +| = \frac{1}{2} (|0\rangle\langle 0| + |0\rangle\langle 1| + |1\rangle\langle 0| + |1\rangle\langle 1|) \doteq \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \\ \rho_{\text{mix}} &= \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) \doteq \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

The difference lies in the off-diagonal elements, which correlate with how coherent the state is. This is manifestly evident in the X -basis measurement, where:

- $P_+(+) = 1$ and $P_+(-) = 0$, but
- $P_{\text{mix}}(+) = P_{\text{mix}}(-) = \frac{1}{2}$.

Problem 9

$$|\psi_A\rangle \otimes |\phi_B\rangle = \left(\frac{|0\rangle_A + |1\rangle_A}{\sqrt{2}} \right) \otimes |1\rangle_B = \frac{1}{\sqrt{2}} (|01\rangle_{AB} + |11\rangle_{AB}) \doteq \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

Problem 10

$$\begin{aligned} \rho &= \frac{1}{2} (I + \langle X \rangle X + \langle Y \rangle Y + \langle Z \rangle Z) \\ &= \frac{1}{2} I + \frac{3}{10} X + \frac{2}{5} Z \\ &= \frac{9}{10} |0\rangle\langle 0| + \frac{3}{10} (|0\rangle\langle 1| + |1\rangle\langle 0|) + \frac{1}{10} |1\rangle\langle 1| \\ &\doteq \begin{pmatrix} \frac{9}{10} & \frac{3}{10} \\ \frac{3}{10} & \frac{1}{10} \end{pmatrix} \end{aligned}$$

Simple calculations yield $\rho^2 = \rho$, meaning the state is pure. In fact, one could infer this from the fact that $\langle X \rangle^2 + \langle Y \rangle^2 + \langle Z \rangle^2 = 1$.

Problem 11

A projective measurement $\{P_m\}$ is defined via the following conditions:

- Each operator is an orthogonal projection: $\forall m. P_m = P_m^2 = P_m^\dagger$
- The ranges of each projection are pairwise orthogonal: $\forall m, n. P_m P_n = \delta_{mn} P_n$
- The projections are collectively complete: $\sum_m P_m = I$

Problem 12

For fun, let us use the density matrix formalism for this problem:

$$\rho = |\psi\rangle\langle\psi| \doteq \begin{pmatrix} \frac{1}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}.$$

The computational basis measurement can be formulated as a PVM:

$$\Pi_0 = |0\rangle\langle 0|, \quad \Pi_1 = |1\rangle\langle 1|.$$

The Born rule allows us to find the probabilities of each measurement outcome:

$$P(0) = \text{Tr}(\Pi_0 \rho) = \frac{1}{4}, \quad P(1) = \text{Tr}(\Pi_1 \rho) = \frac{3}{4}.$$

Problem 13

The post-measurement state can be calculated in the PVM formalism as well:

$$\rho_0 = \frac{\Pi_0 \rho \Pi_0}{P(0)} = |0\rangle\langle 0| \Rightarrow |\psi\rangle_0 = |0\rangle$$

$$\rho_1 = \frac{\Pi_1 \rho \Pi_1}{P(1)} = |1\rangle\langle 1| \Rightarrow |\psi\rangle_1 = |1\rangle$$

Problem 14

A POVM is defined to be Hermitian, positive semi-definite operators that sum to unity. The given operators form a POVM set:

- $E_0^\dagger = E_0$, $E_0 \geq 0$ ($\cdot$: eigenvalues: $\frac{2}{3}, 0$)
- $E_1 = \frac{1}{3} |0\rangle\langle 0| + |1\rangle\langle 1|$, so $E_1^\dagger = E_1$, $E_1 \geq 0$ ($\cdot$: eigenvalues: $\frac{1}{3}, 1$), $E_0 + E_1 = I$

For the state $|0\rangle$, the outcome probabilities are:

$$P(E_0) = \text{Tr}(E_0 |0\rangle\langle 0|) = \langle 0|E_0|0\rangle = \frac{2}{3}, \quad P(E_1) = \text{Tr}(E_1 |0\rangle\langle 0|) = \langle 0|E_1|0\rangle = \frac{1}{3}.$$

Problem 15

Suppose Bob uses some POVM $\{E, I - E\}$ ($0 \leq E \leq I$) to discriminate the states, predicting $|0\rangle$ in the first measurement outcome and $|+\rangle$ in the second. If Alice prepares $|0\rangle$ with probability p and $|+\rangle$ with probability $1 - p$, the success probability is given as

$$P_{\text{success}} = p \langle 0|E|0\rangle + (1 - p) \langle +|(I - E)|+\rangle$$

$$= 1 - p + \text{Tr}((p|0\rangle\langle 0| - (1 - p)|+\rangle\langle +|)E).$$

To maximize the trace in the last expression, E must equal the projection onto the positive subspace of $p|0\rangle\langle 0| + (1 - p)|+\rangle\langle +|$. Even in such a case, however, because $|0\rangle$ and $|+\rangle$ are non-orthogonal, the success probability cannot reach 1 for any E , meaning a perfect state discrimination is impossible.

Problem 16

Alice's measurements are represented by the PVM:

$$\Pi_0 = |0\rangle\langle 0| \otimes I_B, \quad \Pi_1 = |1\rangle\langle 1| \otimes I_B.$$

The probability of either outcome is $\frac{1}{2}$, and the joint state collapses to

$$\begin{cases} |00\rangle & (\text{Outcome: } 0) \\ |11\rangle & (\text{Outcome: } 1). \end{cases}$$

Without classical communication of Alice's measurement outcome, Bob's reduced density matrix is the probabilistic mixture $\rho_B = \frac{1}{2}I_B$, which is the same as before Alice made her measurement. Thus, Alice's measurement did not affect Bob's qubit!

Compared to the joint Bell-state measurement, which can be realized by a coordinated measurement between Alice and Bob and yields perfect correlation, this local measurement differs by the fact that the measurement outcome is not known by Bob. Bob's reduced density matrix is equivalent to a probabilistic mixture, even though the joint state is coherent. Thus, this experiment shows that entanglement is a nonlocal phenomenon that cannot be detected by local operations without any communication.

Problem 17

$$\rho_{AB} = |\Phi^+\rangle\langle\Phi^+| \doteq \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \rho_A = \text{Tr}_B \rho_{AB} \doteq \frac{1}{2} \begin{pmatrix} \text{Tr} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} & \text{Tr} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\ \text{Tr} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} & \text{Tr} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2} I_A$$

Problem 18

The state is already in Schmidt form, as $|0\rangle$ and $|1\rangle$ are orthonormal.

$$\therefore s_0 = \cos \theta, \quad s_1 = \sin \theta$$

Problem 19

$$\rho_A = \cos^2 \theta |0\rangle\langle 0| + \sin^2 \theta |1\rangle\langle 1|$$

$$\Rightarrow E_{AB} = S(\rho_A) = -\cos^2 \theta \log_2 \cos^2 \theta - \sin^2 \theta \log_2 \sin^2 \theta$$

$$(a) E_{AB}(\theta = 0) = 0 \quad (b) E_{AB}(\theta = \frac{\pi}{4}) = 1$$

Problem 20

No, the two qubits are not entangled. One possible description of such an ensemble is an equal probability mixture between $|00\rangle$ and $|11\rangle$, both of which are separable. Thus, there is only a classical correlation, not quantum entanglement. Note that because the state is not pure, a nonzero entanglement entropy (which happens to be 1 for this state) cannot reliably determine entanglement.

Problem 22 [*sic.*]

(Copied from problem 17)

$$\rho_{AB} = |\Phi^+\rangle\langle\Phi^+| \doteq \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow \rho_A = \text{Tr}_B \rho_{AB} = \frac{1}{2} \begin{pmatrix} \text{Tr} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} & \text{Tr} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\ \text{Tr} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} & \text{Tr} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{1}{2} I_A$$